

Shubham Kumar Patel

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025 - Present	M.Sc. Statistics	Indian Institute of Technology, Kanpur	7.12
2023-2025	B.Sc. Statistics	Mahatma Gandhi Kashi Vidyapeeth, Varanasi	8.13
2021	UPMSP(XII)	K M I C Kandawa Purvi, Varanasi	78.4%
2019	UPMSP(X)	K M I C Kandawa Purvi, Varanasi	86.17%

Scholastic Achievements

- Secured **All India Rank 89 in IIT JAM Mathematical Statistics 2025 among the 2880 candidates** across the country.
- Successfully qualified **NIELIT O Level Examination**(Govt. of India) in year 2024, certified in IT tools, Internet Technology, and Web Design.

Key Projects

- Heart Disease Prediction Using Machine Learning** (self project) ([GitHub](#)) (Nov 2025-Dec 2025)
 - Developed a **machine learning model** to predict heart disease using clinical and demographic patient data.
 - Conducted data preprocessing including categorical variable encoding, feature scaling, and train-test splitting.
 - Trained a **Logistic Regression classification model** to identify the presence of heart disease.
 - Evaluated model performance using **accuracy** and **classification metrics** and tested predictions on **new inputs**.
- Future Population Trends: Among the Top Five Countries** (self project) ([GitHub](#)) (Jan 2025-Feb 2025)
 - Analyzed population data (1960-2023) for India, China, USA, Indonesia, and Pakistan using **World Bank datasets**.
 - Used **MS Excel** for model fitting and projection using **Linear, Exponential, and Polynomial Regression**.
 - Applied **least squares method** to estimate parameters and assess model fit using **R² and MSE values**.
 - Identified the model **best fit for each country** and Projected Population growth and trends up to 2040.
- A Statistical Analysis of Sports and Fitness Patterns Based on Survey Data** ([GitHub](#)) (Mar 2025-April 2025)
Mentor: Prof. Raman Pant, Department of Statistics, MGKVP Varanasi.
 - Designed and conducted a structured survey with a sample size of **100 respondents** using convenience sampling.
 - Data collected on age, gender, education, fitness habits, motivation, and use of fitness-related technology.
 - Used **Microsoft Excel** for data cleaning, tabulation, and visualization (bar charts, pie charts, frequency tables).
 - Applied **Chi-square test** to examine the **association between demographic variables and fitness frequency**.

Technical Skills

- Programming Languages:** Python, C, HTML, R, SQL
- Libraries and Packages:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, ggplot2, rvest, dplyr

Teaching and Academic Experience

- Doubt Faculty for IIT JAM Mathematical Statistics at MathStat Classes, handled concept clarification and problem solving, also assisted in proofreading and reviewing IIT JAM Mathematical Statistics mock test papers to ensure mathematical accuracy.

Relevant Courses (* - Ongoing courses)

Probability Theory	Sampling Theory
Descriptive Statistics	Hypothesis Testing
Statistical Inference	Regression Analysis
Complete Data Science, ML, DL, NLP Bootcamp (Udemy)*	Computer Application in Statistics (e.g., R/SPSS practicals)

Extra-Curricular Activities

- Participated in **Annual Sports Day** and cultural events during undergraduate studies.
- Member of the **event organizing committee** for a college-level seminar and function.